

Multi Vector Attack Correlation, SSH Brute Force and Web Reconnaissance

Investigation Date: 14 September 2026

Incident Date: 4 September 2026

Environment: LearnCyber

Primary Host: SOC

Primary SSH Source: 195.178.110.228

Web Attack Source: 195.178.110.15

Assessment: Broader reconnaissance / multi vector malicious activity

Confidence: Medium

Executive Summary

The investigation identified malicious activity targeting multiple LearnCyber attack surfaces. Repeated failed SSH authentication attempts from 195.178.110.228 targeted the root account, while Cloudflare blocked a request from 195.178.110.15 for /wp-config.php-backup. The events occurred approximately three seconds apart, and the investigated IPs shared infrastructure associated with AS48090. The evidence supports broader reconnaissance or multi vector malicious activity, with medium confidence. No successful SSH authentication or successful retrieval of the targeted web resource was identified in the reviewed evidence

Investigation Objective

The objective of this investigation was to determine whether the observed SSH and web based activity represented isolated malicious events or evidence of broader reconnaissance against LearnCyber infrastructure.

The investigation focused on:

1. Triaging the Wazuh SSH alert,
2. Examining Linux authentication logs,
3. Determining whether the SSH activity represented password guessing or brute force behaviour,
4. Performing threat intelligence investigation on related IP addresses,
5. Analysing the Cloudflare security event,

6. Correlating activity across Wazuh, SSH and Cloudflare,
7. Determining whether successful compromise occurred,
8. Mapping observed behaviour to MITRE ATT&CK,
9. Developing containment and remediation recommendations.

Scope

The investigation focused on the following indicators:

Indicator	Purpose
195.178.110.228	Primary SSH source
195.178.110.30	Related threat intelligence investigation
195.178.110.15	Cloudflare web attack source
SOC	Target Linux system
root	Target SSH account
/wp-config.php-back up	Targeted web resource
AS48090	Common ASN identified during threat intelligence analysis
TECHOFF SRV LIMITED	Common infrastructure provider identified

Initial Alert Triage

The initial Wazuh alert contained the following information:

Field	Finding
Source IP	195.178.110.228
Target system	SOC
Target account	root

Authentication method	SSH
Rule description	PAM: User login failed
Rule level	5
Authentication result	Failed
Event timestamp	4 September 2026, 14:30:42

The original investigation recorded the Wazuh rule as level 5 and confirmed that the authentication attempt failed.

SSH Activity Timeline

Time	Source	IP	Activity
14:25	SSH	195.178.110.228	Connection closed
14:28	SSH	195.178.110.228	Root authentication failure
14:29	SSH	195.178.110.228	Failed password for root
14:29	SSH	195.178.110.228	Connection closed by authentication user root
14:30	SSH	195.178.110.228	Root authentication failure
14:30	SSH	195.178.110.228	Failed password for root
14:30	SSH	195.178.110.228	Connection closed by authentication user root
14:32	SSH	195.178.110.228	Root authentication failure
14:32	SSH	195.178.110.228	Failed password for root
14:32	SSH	195.178.110.228	Connection closed by authentication user root

Threat Intelligence Investigation

The investigation examined three IP addresses using VirusTotal, AbuseIPDB, and Shodan as specified in the investigation workflow.

IP Address, 195.178.110.228

The investigation identified the following indicators:

Source	Finding
VirusTotal	16 of 89 vendors flagged the IP as malicious
AbuseIPDB	47,632 reports
Abuse confidence	100%
ASN	AS48090
Provider	TECHOFF SRV LIMITED
Country shown by AbuseIPDB / Shodan	Netherlands
Scanner classification	Brute force, hacking, exploited host, SSH

The evidence supports the assessment that 195.178.110.228 has a significant history of malicious or abusive activity.

IP Address, 195.178.110.30

The second IP demonstrated similar infrastructure characteristics:

Source	Finding
VirusTotal	12 of 89 vendors flagged the IP as malicious
AbuseIPDB	412,849 reports, as recorded in the investigation
Abuse confidence	100%
ASN	AS48090
Provider	TECHOFF SRV LIMITED

Country shown by AbuseIPDB / Shodan	Netherlands
Scanner classification	Brute force, SSH

IP Address, 195.178.110.15

The third IP demonstrated:

Source	Finding
VirusTotal	11 of 89 vendors flagged the IP as malicious
AbuseIPDB	178,133 reports
Abuse confidence	100%
ASN	AS48090
Provider	TECHOFF SRV LIMITED
Country shown by AbuseIPDB / Shodan	Netherlands
Classifications	Brute force, hacking, web application attack, port scan, DDoS, Bad Web Bot, exploited host, SSH

Threat intelligence interpretation

The three investigated IPs shared significant infrastructure characteristics, particularly:

- AS48090
- TECHOFF SRV LIMITED

The country information also differed between sources in VirusTotal the geolocation identified is Bulgaria which on Shodan the geolocation identified is Netherlands

MITRE ATT&CK Mapping

T1110.001, Password Guessing

Status: Confirmed

Evidence:

Repeated SSH password authentication failures against the root account were observed in the Linux authentication logs.

This is the strongest MITRE ATT&CK mapping in the investigation.

T1595, Active Scanning

Status: Potentially applicable

Threat intelligence associated 195.178.110.15 with scanning related activity, including port scanning.

However, threat intelligence classification alone does not prove that active scanning occurred against LearnCyber.

Therefore: T1595 is considered potentially applicable, but additional LearnCyber telemetry would be required to confirm active scanning against the organisation.

T1190, Exploit Public Facing Application

Status: Potential, not confirmed

The Cloudflare event involved a request for: /wp-config.php-backup

The request was blocked by a Cloudflare rule associated with broken access control and file inclusion.

However, requesting a potentially sensitive file does not by itself establish successful exploitation.

Therefore, T1190 should not be presented as a confirmed technique.

Containment Recommendations

Immediate Actions

1. Confirm that SSH password guessing is no longer ongoing
2. Confirm successful authentication status

3. Review privileged account activity
4. Block confirmed malicious IP addresses
5. Continue Cloudflare monitoring

Short Term Recommendations

SSH hardening

1. Disable direct root SSH login where operationally possible.
2. Prefer SSH keys over password authentication.
3. Restrict SSH access to trusted networks, VPN or approved administrative sources where possible.
4. Review firewall rules and UFW configuration.
5. Implement rate limiting or automated blocking for repeated SSH authentication failures.

Web security

1. Confirm that configuration backups cannot be downloaded from the public web.
2. Search the web server for exposed backup copies of sensitive configuration files.
3. Review Cloudflare WAF events for similar requests.
4. Confirm that the origin cannot be directly accessed in a way that bypasses Cloudflare protections.

Long Term Recommendations

The investigation identified an opportunity to improve correlation between security controls.

Recommended improvements include:

1. Develop Wazuh rules for repeated authentication failures against privileged accounts.
2. Correlate SSH authentication events with source IP reputation.
3. Forward relevant Cloudflare security telemetry into the SIEM.
4. Create correlation rules for simultaneous web and infrastructure attacks.
5. Enrich security alerts automatically with threat intelligence.
6. Monitor for repeated activity from infrastructure associated with previously malicious sources.
7. Establish a documented incident response procedure for multi vector attacks.

Evidence Table

Evidence ID	Source	Evidence	Significance
E01	Wazuh	Failed PAM login alert from 195.178.110.228	Identifies suspicious SSH authentication
E02	Linux auth.log	Multiple authentication events from 195.178.110.228	Confirms activity at host level
E03	Linux auth.log	Repeated Failed password for root	Supports password guessing
E04	Wazuh	Event at 14:30:42	Provides correlation timestamp
E05	Cloudflare	Event at 14:30:45	Web activity occurred three seconds later
E06	Cloudflare	GET /wp-config.php-backup	Indicates attempted access to potentially sensitive resource
E07	Cloudflare	Action: Block	Indicates defensive control prevented request
E08	Cloudflare	Rule: Wordpress, Broken Access Control, File Inclusion	Identifies WAF detection
E09	Threat Intelligence	195.178.110.228 malicious reputation	Supports source risk assessment
E010	Threat Intelligence	195.178.110.15 malicious reputation	Supports source risk assessment
E011	Threat Intelligence	Shared AS48090 infrastructure	Supports cross source correlation
E012	Linux authentication logs	No successful SSH authentication identified	No confirmed SSH compromise

Final Conclusion

The investigation identified malicious activity targeting multiple LearnCyber attack surfaces.

The first activity involved repeated failed SSH authentication attempts from 195.178.110.228 targeting the privileged root account on the SOC system. The authentication logs support a classification of repeated SSH password guessing, with no successful SSH authentication identified in the reviewed evidence.

A separate Cloudflare event involving 195.178.110.15 occurred approximately three seconds after the Wazuh SSH event. The source attempted to access /wp-config.php-backup, a resource that could potentially expose sensitive WordPress configuration information if publicly accessible. Cloudflare blocked the request before it reached the origin.

Threat intelligence investigation identified significant malicious reputation associated with the investigated IP addresses and common infrastructure through AS48090 and TECHOFF SRV LIMITED.

Taken together, the evidence supports the assessment that the activity represents broader reconnaissance or multi vector malicious activity against LearnCyber, rather than an isolated SSH event.

The assessment is assigned **Medium confidence** because the temporal, behavioural and infrastructure correlations are meaningful, but the available evidence does not conclusively establish that all activity originated from the same threat actor.

Final Analyst Verdict:

Broader reconnaissance / multi vector malicious activity against LearnCyber, Medium confidence. No successful SSH compromise or successful retrieval of the targeted WordPress backup file was identified in the reviewed evidence.